Week 1 Progress Report

This week marked the official start of the project. I attended my first supervision meeting on Friday 9th October where I presented the project concept and received detailed feedback that will shape the direction of the work going forward.

Prior to the meeting I refined the project scope narrowing from a vague 100+ feature set down to a focused 35 features across 8 distinct, verifiable data sources (EIA, OPEC, CFTC, FRED, Baker Hughes, GDELT, GPR Index, and Brent futures). I also identified the core novel contribution: an LLM-powered geopolitical scoring pipeline using GDELT news data to generate structured monthly risk scores for major oil-producing countries and key maritime chokepoints.

The supervisor highlighted several areas to prioritise. First, the literature survey needs to be engaged with from the start not treated later. Second, model selection requires proper academic justification; citing familiarity with Random Forest, XGBoost and LightGBM is not sufficient and I need to ground each choice in the existing literature. Third, the methodology needs to clearly address which features have the most influence on the outcome SHAP values were discussed as the appropriate tool for this. The overarching goal is to build a convincing case for the examiners as to why this approach works.

Next steps: obtain and verify GDELT data, begin literature review, and draft the project plan structure ahead of the 29 October formative deadline.